

A Cross-Model Evaluation of LLM Output Reconstruction from Token-Length Traces in Controlled SSE Streaming

Author One,  *Member, IEEE*, and Author Two,  *Senior Member, IEEE*

Abstract—Large-language-model (LLM) assistants commonly stream responses token by token using Server-Sent Events (SSE). Frame sizes that remain observable under encryption can correlate with token lengths and thereby leak information about the generated text. This paper reports a *controlled* cross-model evaluation of output reconstruction from *idealized* one-token-per-frame length traces across seven open-weight language models (six instruction/chat-tuned models and one base model; 300 aligned prompts each; 2100 samples). A shared Weiss-style T5 pipeline recovers text hypotheses from length sequences. We score reconstructions with a single fixed *term-frequency cosine similarity* ϕ (a lexical overlap measure, not a paraphrase-aware semantic embedding) and report the lexical recovery rate LR@0.5 as the fraction of samples with $\phi > 0.5$. Beyond point estimates we report bootstrap 95% confidence intervals and paired Wilcoxon tests with Holm–Bonferroni correction. Mean ϕ varies significantly across models—from 0.324 (Qwen2.5-1.5B-Instruct, 95% CI [0.310, 0.338]) to 0.024 (Phi-3.5-mini-instruct)—and 19 of 21 pairwise differences remain significant after correction. We interpret the ranking as transferability of a fixed reconstructor onto each model’s traces, report response-length covariates, and show via a linear model that model identity retains explanatory power after controlling for length and truncation. Trace-information baselines (shuffled, constant, and cross-prompt lengths) quantify how much signal the real length sequence contributes beyond the reconstructor’s language prior. We further evaluate five transport-preserving defenses (four mechanism classes: bucketing, fixed framing, batching, randomized padding) *against the fixed, non-adaptive reconstructor*, reporting residual ϕ , Wilson intervals for LR@0.5, and byte overhead—not only frame counts. No defended sample exceeded $\phi = 0.5$ in the evaluated subset ($n = 30$ prompts $\times$ 2 models; Wilson upper bounds remain nontrivial). End-to-end TLS extraction and adaptive attackers trained on defended traces are left to future work.

Index Terms—Large language models, side-channel analysis, server-sent events, traffic analysis, privacy leakage, token streaming, text reconstruction, statistical significance, defenses.

I. INTRODUCTION

INTERACTIVE LLM assistants return answers incrementally to reduce perceived latency. A frequent pattern is token streaming over SSE, where each chunk is encapsulated in a frame transmitted over an encrypted transport. Encryption protects payload *content*, but observable sizes can correlate with token lengths [1].

Manuscript submitted July 21, 2026. (*Corresponding author: Author One.*)
The authors are with the Department/Institute, University/Organization, City, Country (e-mail: author1@example.com; author2@example.com).

Code, per-sample metrics, and a SHA-256 artifact manifest accompany this paper (anonymous review mirror: <https://anonymous.4open.science/r/llm-akm-repro-TBD>).

Weiss *et al.* demonstrated token-length reconstruction on live ChatGPT and Copilot traffic (27% response recovery; 53% topic inference under their criteria) [1]. Concurrent work studies merged API outputs [2] and related streaming metadata attacks [3]. We ask a complementary question: under a controlled, idealized one-token-per-frame observation model, how does reconstruction quality vary across open-weight models when prompts, decoding, reconstructor, and metrics are held fixed?

Scope. We study an *ideal token-length oracle* from an instrumented harness, not TLS ciphertext extraction. This provides an *optimistic observation model* for a length-only adversary; bridging to live encrypted traffic remains future work (Section III).

Contributions.

- 1) A controlled cross-model reconstruction benchmark on aligned traces from seven open-weight models, with a fixed lexical scoring backend (Sections IV–V).
- 2) Uncertainty-aware claims: bootstrap CIs, paired Wilcoxon tests with Holm correction, effect sizes, length covariates, and a regression controlling for those covariates.
- 3) Trace-information baselines (shuffled / constant / cross-prompt) separating length-channel signal from reconstructor priors (Section VI).
- 4) Defense evaluation of five configurations against a *fixed non-adaptive* reconstructor, with byte overhead and Wilson intervals for LR@0.5 (Section VII).
- 5) A reproducibility package with hashed artifacts (Section IX).

II. RELATED WORK

Side-channel attacks exploit timing, size, and related observables [4]–[6]. Website fingerprinting and traffic-morphing defenses motivate size-domain mitigations [7]–[10]. Weiss *et al.* introduced LLM token-length reconstruction with known-plaintext fine-tuning on live traffic [1]. Li *et al.* study merged token packets [2]. Our focus is a controlled multi-model benchmark under idealized traces, with statistical testing, trace baselines, and a non-adaptive defense sweep—not a new remote exploit.

III. THREAT MODEL AND EXPERIMENTAL SCOPE

We study an adversary who obtains per-token UTF-8 lengths $L_1^{\text{tok}}, \dots, L_n^{\text{tok}}$ and reconstructs the response (Fig. 1). In

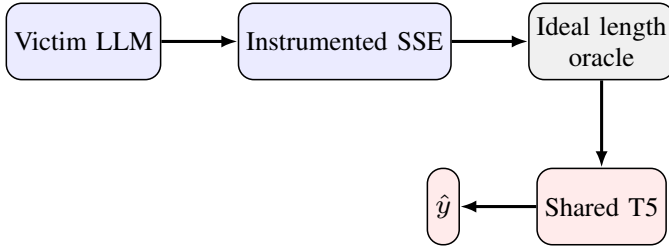


Fig. 1. Ideal length oracle from an instrumented harness (not TLS capture).

Algorithm 1 Weiss-style reconstruction from length traces**Require:** lengths $L = (L_1^{\text{tok}}, \dots, L_n^{\text{tok}})$

- 1: $S \leftarrow \text{SEGMENTSENTENCES}(L)$ ▷ rules above
- 2: decode first segment with T5-First; middle segments with T5-Middle
- 3: **return** concatenated top-ranked hypothesis $\hat{y}$

our harness, $L_t^{\text{tok}} = L_t^{\text{frame}} - 8$ for SSE frames data: `<token>\n\n`. Reconstruction uses only the length sequence.

This matches the token-length *channel abstraction* of Weiss *et al.* [1], but lengths come from an instrumented one-token-per-frame server. A network adversary would still need to recover comparable lengths from TLS/HTTP/2/QUIC records that may coalesce, fragment, or pad frames. We therefore claim reconstructability under an idealized oracle, which is optimistic relative to ciphertext-only observation.

IV. MATERIALS AND METHODS

A. Harness and Trace Collection

The harness emits one token per SSE frame and logs token UTF-8 length, frame length, and timestamps. Token text is retained only for auditing and offline metric recomputation.

B. Reconstruction Pipeline

We use `royweiss1/T5_FirstSentences` and `royweiss1/T5_MiddleSentences` without per-model fine-tuning (Algorithm 1). Candidates are ranked by the unnormalized sum of transition log-probabilities (shorter candidates can be favoured). Cross-model differences therefore measure *transfer* of this fixed reconstructor [1].

Segmentation rules (implemented in `weiss_reconstruction.py`): (1) split after every length-1 token (treated as punctuation under the Weiss heuristic); (2) move a trailing (3,1,1) list marker onto the next segment; (3) merge any segment with fewer than 10 tokens into the following segment; (4) the final remainder forms its own segment. A length-1 token is not guaranteed to be sentence-final across tokenizers; the rule is part of the fixed attack configuration.

C. Models and Protocol

Seven open-weight models: six instruction/chat-tuned and one base model (Phi-1.5). Primary comparative claims use the six instruction/chat models; Phi-1.5 is retained as a stress

TABLE I
MODEL METADATA. KIND: INSTRUCT/CHAT VS. BASE.

Model	Params	Kind	Tokenizer family
Qwen2.5-1.5B-Instruct	~1.5B	instruct	Qwen2
Qwen2.5-3B-Instruct	~3B	instruct	Qwen2
Llama-3.2-3B-Instruct	~3B	instruct	Llama
Gemma-2-2B-it	~2B	instruct	Gemma
Phi-1.5	~1.3B	base	CodeGen
TinyLlama-1.1B-Chat	~1.1B	chat	Llama
Phi-3.5-mini-instruct	~3.8B	instruct	Llama

TABLE II
CONTROL COVARIATES ($N = 300$). TRUNC. %: RESPONSES WITH ≥ 96 TOKENS.

Model	Par.	Kind	Mean tok	Med.	Chars	Mean B/tol
Qwen2.5-1.5B-Instruct	~1.5B	instruct	92.2	96	464	5.03
Qwen2.5-3B-Instruct	~3B	instruct	96.0	96	482	5.03
Llama-3.2-3B-Instruct	~3B	instruct	94.7	96	463	4.88
Gemma-2-2B-it	~2B	instruct	96.0	96	435	4.53
Phi-1.5	~1.3B	base	96.0	96	497	5.17
TinyLlama-1.1B-Chat	~1.1B	chat	96.0	96	331	3.45
Phi-3.5-mini-instruct	~3.8B	instruct	96.0	96	342	3.57

test (Table I). Shared settings: $N = 300$ aligned prompts, temperature 0, top- $p = 1$, max-new-tokens 96, three samples per segment, and at most three sentences. Truncation rates of 95–100% mean nearly all answers are cut mid-stream; results characterize early-response reconstructability under a tight budget.

D. Metrics: Lexical Cosine ϕ and LR@0.5

We recompute for every sample a *term-frequency cosine similarity* ϕ between bag-of-words vectors of the reference and the hypothesis ($\uparrow$). This measures lexical overlap; it is *not* Sentence-BERT or BERTScore and can assign high scores to lexically similar but semantically wrong text (and vice versa). We avoid calling ϕ “semantic similarity.”

The lexical recovery rate LR@0.5 ($\uparrow$) is the fraction of samples with $\phi > 0.5$ (Wilson CI). The threshold is conventional, not human-calibrated. We also report ed_{norm} ($\downarrow$) and ROUGE-1/L precision ($\uparrow$) [11], [12].

E. Statistics

Bootstrap 95% CIs ($B = 10,000$) for means; Wilson intervals for LR@0.5; paired Wilcoxon tests on per-prompt ϕ with Holm–Bonferroni over $\binom{7}{2} = 21$ pairs and rank-biserial r [13]–[15]. Seed: 20260706.

V. RESULTS

A. Control Covariates

Table II shows mean/median token counts, characters, mean token UTF-8 bytes, and truncation. Equal token budgets do not imply equal text across tokenizers.

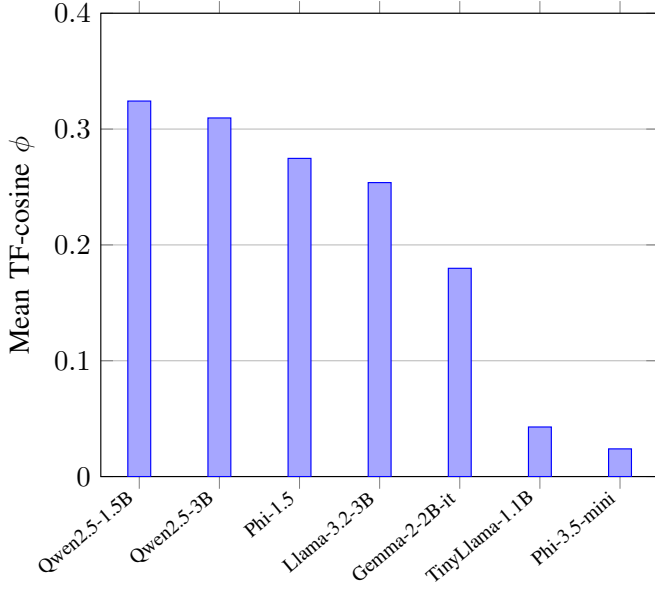


Fig. 2. Mean TF-cosine ϕ with 95% bootstrap CIs ($N = 300$).

B. Cross-Model Comparison

Table III and Fig. 2 report ϕ with CIs (all seven models). Among instruction/chat models only (Table IV), the same ordering of extremes holds. Phi-1.5 ranks high under lexical cosine despite being a base model—consistent with reconstructor transfer, not chat alignment.

C. Significance and Covariate Control

Table V: 19/21 pairs significant after Holm correction. A linear model $\phi = \beta_0 + \beta_{\text{model}} + \beta_1(\text{chars}/100) + \beta_2|t| + \beta_3\text{trunc}$ on the six instruction/chat models yields $R^2 = 0.646$, versus 0.642 for model indicators alone and 0.511 for covariates alone (Table VI). Model identity retains nearly all explanatory power after length controls; covariates alone are insufficient to explain the ranking.

VI. TRACE-INFORMATION BASELINES

To isolate the contribution of the length channel beyond the T5 language prior, we reconstruct from (i) the real length sequence, (ii) a within-response shuffle, (iii) a constant sequence at the mean token length, and (iv) a cross-prompt length sequence of equal cardinality, on the same topic-balanced subset used for defenses ($n = 30$ prompts per model). Table VII shows that real traces outperform all degraded traces: shuffling lowers mean ϕ by about 0.09–0.10, and a constant-length sequence lowers it by about 0.17–0.20. Cross-prompt lengths remain closer to real ($\Delta \approx -0.05$ to -0.06), indicating that response length *structure*—not only marginal length statistics—carries usable information under the fixed reconstructor.

VII. DEFENSE EVALUATION

A. Defenses (Non-Adaptive Attacker)

We evaluate *five* defense configurations (four classes) that transform observable lengths before reconstruction with the

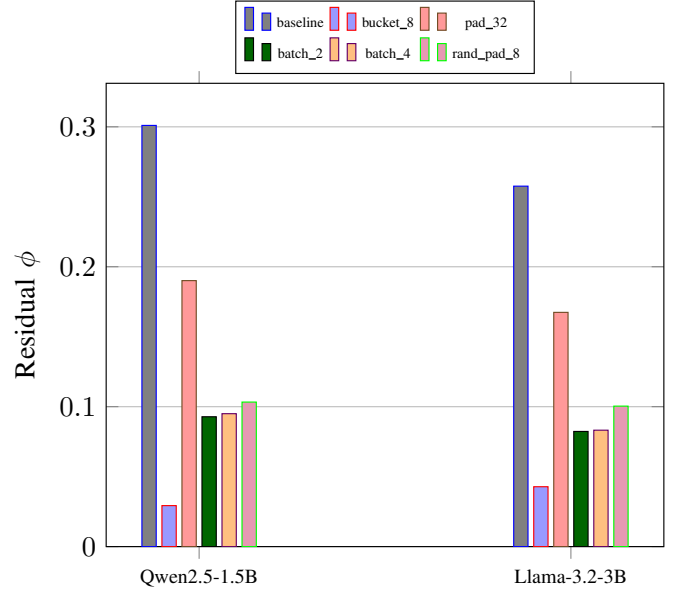


Fig. 3. Residual ϕ (mean $\pm$ 95% CI) under defenses (fixed reconstructor).

same undefended-trained T5 pipeline—i.e., a fixed, non-adaptive attacker who does not retrain on defended traces [1], [10]:

- `bucket_8`: ceil each length to a multiple of 8;
- `pad_32`: online fixed frame size $C = 32$ (no response-wide look-ahead; tokens longer than C still emit size C);
- `batch_2` / `batch_4`: one frame per k tokens;
- `rand_pad_8`: uniform padding in $\{0, \dots, 8\}$ (seeded).

We do *not* claim security against an adaptive adversary who fine-tunes on padded/batched traces.

B. Results

Table VIII reports residual ϕ , LR@0.5 with Wilson intervals, mean byte overhead $\text{OH} = 100(\text{bytes}_{\text{def}} - \text{bytes}_{\text{base}})/\text{bytes}_{\text{base}}$, and leakage reduction vs. each model’s own baseline ($n = 30$ topic-balanced prompts; Qwen2.5-1.5B and Llama-3.2-3B). No defended sample exceeded $\phi = 0.5$ in this subset; Wilson upper bounds are about 11%, so “zero observed successes” must not be read as a point estimate of zero risk. Bucketing is the strongest among the tested options against the frozen reconstructor (≈ 83 – 90% ϕ reduction). Online `pad_32` is weaker (≈ 35 – 37% reduction) despite very large byte overhead (≈ 5 – $6\times$), because a constant frame size still admits a trivial reconstruction prior when C is far above typical token UTF-8 lengths; smaller online constants (e.g., $C \in \{8, 16\}$) are a natural follow-up. Batching reduces frame count at zero byte sum overhead in our accounting and yields $\approx 68\%$ reduction; randomized padding sits in between.

VIII. QUALITATIVE EXAMPLES

Table IX illustrates high and low TF-cosine cases. High ϕ can preserve boilerplate structure while altering key entities (career vs. barrister), underscoring that LR@0.5 is a lexical—not semantic—proxy.

TABLE III
ALL MODELS ($N = 300$). ϕ : TF COSINE ($\uparrow$); ed_{norm} ($\downarrow$); LR@0.5 ($\uparrow$, %).

Model	$\phi \uparrow$	95% CI	$ed \downarrow$	R1 $\uparrow$	RL $\uparrow$	LR@0.5
Qwen2.5-1.5B-Instruct	0.3242	[0.3097, 0.3380]	0.7199	0.3220	0.2651	8.67 [5.98, 12.40]
Qwen2.5-3B-Instruct	0.3096	[0.2977, 0.3214]	0.7300	0.2945	0.2345	3.00 [1.59, 5.60]
Phi-1.5	0.2747	[0.2622, 0.2877]	0.7426	0.2503	0.1891	4.33 [2.55, 7.27]
Llama-3.2-3B-Instruct	0.2538	[0.2419, 0.2658]	0.7499	0.2387	0.1824	3.00 [1.59, 5.60]
Gemma-2-2B-it	0.1798	[0.1708, 0.1889]	0.7764	0.1719	0.1269	0.00 [0.00, 1.26]
TinyLlama-1.1B-Chat	0.0428	[0.0377, 0.0479]	0.8026	0.0312	0.0310	0.00 [0.00, 1.26]
Phi-3.5-mini-instruct	0.0239	[0.0201, 0.0280]	0.8087	0.0158	0.0157	0.00 [0.00, 1.26]

TABLE IV
INSTRUCTION/CHAT MODELS ONLY (PHI-1.5 EXCLUDED).

Model	$\phi \uparrow$	95% CI	$ed \downarrow$	R1 $\uparrow$
Qwen2.5-1.5B-Instruct	0.3242	[0.3097, 0.3380]	0.7199	0.3220
Qwen2.5-3B-Instruct	0.3096	[0.2977, 0.3214]	0.7300	0.2945
Llama-3.2-3B-Instruct	0.2538	[0.2419, 0.2658]	0.7499	0.2387
Gemma-2-2B-it	0.1798	[0.1708, 0.1889]	0.7764	0.1719
TinyLlama-1.1B-Chat	0.0428	[0.0377, 0.0479]	0.8026	0.0312
Phi-3.5-mini-instruct	0.0239	[0.0201, 0.0280]	0.8087	0.0158

APPENDIX

Best/worst topics ($N = 20/\text{topic}$) for three models are exploratory only; full means are in `topic_matrix.json`. No interaction test is claimed.
Python distribution/package versions: see `environment.yml` and `requirements.txt`.
Data format/package versions: see `data_format.json` (numpy, torch, transformers).
Decoding: deterministic; chat templates follow each model's Hugging Face defaults.

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IX. REPRODUCIBILITY

Scripts: `analyze_stats.py`, `analyze_extras.py`, `defense_eval.py`, `trace_baselines.py`, `make_figures.py`, `make_repro_manifest.py`. Environment snapshot: `env_snapshot.json`. Artifact hashes: `repro_manifest.json`. Tables/figures are \input from `experiment_validation/analysis/`.

X. DISCUSSION

Mean reconstructability under idealized traces varies significantly across the evaluated models. The ranking is reconstructor transfer, not intrinsic leakage after per-model adaptation. Defenses substantially reduced observed ϕ against the fixed reconstructor and yielded no LR@0.5 successes in the pilot subset; they are *promising mitigations against that attacker model*, not proven robust to adaptive retraining. High truncation under 96 tokens limits ecological validity; sensitivity runs at 256–384 tokens are recommended.

XI. CONCLUSION

We presented a controlled cross-model evaluation of LLM output reconstruction from idealized SSE length traces, with lexical (not embedding) scoring, paired statistics, covariate analysis, trace baselines, and a five-configuration defense study against a non-adaptive reconstructor. Future work: live TLS extraction, adaptive defended-trace attackers, Sentence-BERT/BERTScore with human ratings, longer generation budgets, and multi-seed randomized padding.

ACKNOWLEDGMENT

We thank the maintainers of the open models and the `royweiss1` checkpoints.

TABLE V
 PAIRED WILCOXON TESTS ON ϕ ($n = 300$). *: $p_{\text{Holm}} < 0.05$.

Comparison	$\Delta\phi$	95% CI	r	p_{raw}	p_{Holm}
Qwen2.5-1.5B-Instruct vs. Gemma-2-2B-it	+0.1443	[+0.1276, +0.1608]	+0.837	<0.0001	<0.0001*
Qwen2.5-1.5B-Instruct vs. TinyLlama-1.1B-Chat	+0.2814	[+0.2660, +0.2967]	+0.999	<0.0001	<0.0001*
Qwen2.5-1.5B-Instruct vs. Phi-3.5-mini-instruct	+0.3002	[+0.2851, +0.3152]	+1.000	<0.0001	<0.0001*
Qwen2.5-3B-Instruct vs. Gemma-2-2B-it	+0.1297	[+0.1145, +0.1450]	+0.826	<0.0001	<0.0001*
Qwen2.5-3B-Instruct vs. TinyLlama-1.1B-Chat	+0.2668	[+0.2542, +0.2795]	+1.000	<0.0001	<0.0001*
Qwen2.5-3B-Instruct vs. Phi-3.5-mini-instruct	+0.2856	[+0.2727, +0.2985]	+1.000	<0.0001	<0.0001*
Llama-3.2-3B-Instruct vs. Gemma-2-2B-it	+0.0739	[+0.0595, +0.0889]	+0.566	<0.0001	<0.0001*
Llama-3.2-3B-Instruct vs. TinyLlama-1.1B-Chat	+0.2110	[+0.1978, +0.2244]	+0.994	<0.0001	<0.0001*
Llama-3.2-3B-Instruct vs. Phi-3.5-mini-instruct	+0.2298	[+0.2168, +0.2430]	+0.998	<0.0001	<0.0001*
Gemma-2-2B-it vs. Phi-1.5	-0.0948	[-0.1105, -0.0792]	-0.681	<0.0001	<0.0001*
Gemma-2-2B-it vs. TinyLlama-1.1B-Chat	+0.1370	[+0.1268, +0.1472]	+0.977	<0.0001	<0.0001*
Gemma-2-2B-it vs. Phi-3.5-mini-instruct	+0.1559	[+0.1457, +0.1663]	+0.990	<0.0001	<0.0001*
Phi-1.5 vs. TinyLlama-1.1B-Chat	+0.2319	[+0.2179, +0.2459]	+0.998	<0.0001	<0.0001*
Phi-1.5 vs. Phi-3.5-mini-instruct	+0.2507	[+0.2367, +0.2649]	+0.999	<0.0001	<0.0001*
Qwen2.5-1.5B-Instruct vs. Llama-3.2-3B-Instruct	+0.0704	[+0.0530, +0.0875]	+0.498	<0.0001	<0.0001*
Qwen2.5-3B-Instruct vs. Llama-3.2-3B-Instruct	+0.0558	[+0.0394, +0.0719]	+0.426	<0.0001	<0.0001*
TinyLlama-1.1B-Chat vs. Phi-3.5-mini-instruct	+0.0189	[+0.0132, +0.0245]	+0.454	<0.0001	<0.0001*
Qwen2.5-1.5B-Instruct vs. Phi-1.5	+0.0495	[+0.0305, +0.0686]	+0.330	<0.0001	<0.0001*
Qwen2.5-3B-Instruct vs. Phi-1.5	+0.0349	[+0.0165, +0.0528]	+0.268	<0.0001	0.0002*
Llama-3.2-3B-Instruct vs. Phi-1.5	-0.0209	[-0.0375, -0.0043]	-0.148	0.0266	0.0533
Qwen2.5-1.5B-Instruct vs. Qwen2.5-3B-Instruct	+0.0146	[-0.0026, +0.0314]	+0.089	0.1838	0.1838

TABLE VI
 OLS FIT FOR ϕ (INSTRUCT/CHAT MODELS, $n = 1800$).

Quantity	Value
Full model R^2	0.646
Model indicators only R^2	0.642
Covariates only (chars, token bytes, trunc.) R^2	0.511
$\beta_{\text{chars}/100}$	-0.0179
$\beta_{\text{mean token bytes}}$	+0.0290
$\beta_{\text{truncated}}$	-0.0295

TABLE VII
 TRACE-INFORMATION BASELINES (TF-COSINE ϕ). NEGATIVE Δ VS. REAL
 MEANS THE DEGRADED TRACE YIELDS LOWER LEXICAL OVERLAP.

Model	Condition	n	ϕ	95% CI	Δ vs. real
Qwen2.5-1.5B-Instruct	real	30	0.3024	[0.2607, 0.3457]	—
	shuffled	30	0.2091	[0.1734, 0.2463]	-0.0933
	constant	30	0.1068	[0.0844, 0.1303]	-0.1957
	cross_prompt	30	0.2561	[0.2296, 0.2837]	-0.0463
Llama-3.2-3B-Instruct	real	30	0.2913	[0.2568, 0.3282]	—
	shuffled	30	0.1872	[0.1623, 0.2119]	-0.1041
	constant	30	0.1242	[0.0981, 0.1529]	-0.1671
	cross_prompt	30	0.2312	[0.2010, 0.2618]	-0.0601

TABLE VIII
DEFENSES VS. FIXED RECONSTRUCTOR ($n = 30/\text{MODEL}$). LR@0.5 WITH WILSON 95% CI; BYTE OVERHEAD %; LEAK. REDUCTION %. ONLINE FIXED FRAMING USES $C = 32$.

Model	Def.	n	ϕ	95% CI	LR@0.5	Byte OH%	Red.%
Qwen2.5-1.5B-Instruct	baseline	30	0.3010	[0.2537, 0.3508]	13.3 [5.3, 29.7]	0.0	0.0
	bucket_8	30	0.0293	[0.0198, 0.0397]	0.0 [0.0, 11.4]	82.4	90.3
	pad_32	30	0.1901	[0.1667, 0.2150]	0.0 [0.0, 11.4]	538.0	36.8
	batch_2	30	0.0928	[0.0791, 0.1072]	0.0 [0.0, 11.4]	0.0	69.2
	batch_4	30	0.0950	[0.0668, 0.1269]	0.0 [0.0, 11.4]	0.0	68.4
	rand_pad_8	30	0.1033	[0.0871, 0.1203]	0.0 [0.0, 11.4]	81.6	65.7
Llama-3.2-3B-Instruct	baseline	30	0.2576	[0.2177, 0.2978]	3.3 [0.6, 16.7]	0.0	0.0
	bucket_8	30	0.0428	[0.0305, 0.0563]	0.0 [0.0, 11.4]	88.6	83.4
	pad_32	30	0.1674	[0.1382, 0.2003]	0.0 [0.0, 11.4]	565.1	35.0
	batch_2	30	0.0823	[0.0684, 0.0967]	0.0 [0.0, 11.4]	0.0	68.0
	batch_4	30	0.0832	[0.0615, 0.1055]	0.0 [0.0, 11.4]	0.0	67.7
	rand_pad_8	30	0.1004	[0.0834, 0.1186]	0.0 [0.0, 11.4]	85.0	61.0

Effectiveness is against the frozen baseline reconstructor only.

TABLE IX
ILLUSTRATIVE RECONSTRUCTIONS (TRUNCATED). FULL STRINGS IN `QUALITATIVE_EXAMPLES.JSON`.

Model	ϕ	Reference (excerpt)	Hypothesis (excerpt)
Qwen 1.5B	0.69	disclose a disability at work...keep a journal	become a barrister in Ohio...keep a journal
Qwen 1.5B	0.08	managing credit card debt...budget plan	electric guitars...advertising

TABLE X
EXPLORATORY BEST/WORST TOPICS BY MEAN ϕ .

Model	Best	ϕ	Worst	ϕ
Llama-3.2-3B-Instruct	employment	0.3373	safety_security	0.2053
Gemma-2-2B-it	beliefs_values	0.2090	financial_status	0.1575
Phi-3.5-mini-instruct	financial_status	0.0444	personal_identity	0.0073